

# SEBA-XAI: Explainable Policy-Aware Audit for Secure Inter-Agency Police Data Access Governance

Alapati Venkat Rayudu

**Abstract**—Inter-agency police data sharing is not only a database-sharing problem. Sensitive record access needs contextual authorization, tamper-evident audit, and explanations that can be reviewed later. This paper proposes SEBA-XAI, an application-layer access-governance and audit overlay for CCTNS/ICJS-compatible police record workflows. Blockchain is used because multiple agencies may need a shared tamper-evident audit trail without copying raw records into one central database. The system keeps sensitive records off-chain and combines an ABAC-based policy service, permissioned ledger-backed audit records, and an explainable decision service for allow, deny, and escalate requests. The paper also studies a post-decision corruption case: a compromised component changes the recorded policy decision and re-signs the event, so the audit log still appears cryptographically valid. Because real police access logs are not public, the evaluation uses a reproducible synthetic workload. The results show that integrity-only audit baselines detect ordinary tampering but miss the validly re-signed corruption case. In the same benchmark, NS-PI drift detection and a trusted raw-attribute policy oracle detect that case under different visibility assumptions. These findings support a scoped prototype claim: access control, blockchain audit, and explainable review artifacts address different failure modes and should be evaluated together.

**Index Terms**—Access control, blockchain audit, explainable AI, police data governance, CCTNS, ICJS, ABAC, XAI, cybersecurity.

## I. Introduction

Police data sharing is often discussed as an infrastructure problem. This paper treats it as an access-governance problem. India already has major digital policing and criminal-justice infrastructure. The Crime and Criminal Tracking Network and Systems (CCTNS) is reported by official sources to be in use across all 17,798 police stations as of 1 February 2026 [1]. The Inter-Operable Criminal Justice System (ICJS) is designed to support data exchange between the CCTNS police pillar and other criminal-justice pillars, including courts, prisons, forensic services, and prosecution workflows [2]. Because of this, the research question is not whether India needs a new replacement for CCTNS or ICJS. The question is more specific: if sensitive records move through an inter-agency workflow, how can access decisions be controlled, audited, and explained in a way that can be reviewed later? Although India provides the application context for this paper, the underlying problem is not limited to India. Recent international work studies accountable blockchain-based access control [3], privacy-preserving ABAC [4], blockchain protection for digital evidence [5], AI-related data protection in criminal justice [6], and explainable

AI in law enforcement [7]. These works show that access accountability, privacy, evidentiary integrity, and explanation are current research problems, but they are often studied separately rather than as one access-governance workflow.

This question matters because police records are not ordinary database rows. A single request may involve FIR details, victim or witness information, juvenile records, forensic material, cybercrime complaints, case diary content, evidence references, or court/prosecution-linked documents. A broad role such as “police officer” is not enough to decide access. The same officer may be allowed to view one record, denied another, and escalated for a third depending on jurisdiction, case assignment, purpose, sensitivity level, credential status, approval state, emergency context, and the agency relationship behind the request. A real-world policing example shows why even police metadata must be handled carefully: in 2023, a Police Service of Northern Ireland data breach exposed officer and staff details such as names, ranks, and locations, and the UK Information Commissioner’s Office later imposed a fine over the incident [8], [9]. This example is outside India, but it illustrates the same governance risk: sensitive police information and access metadata can create safety, privacy, and accountability problems if they are not controlled and reviewable.

The intended users in this workflow are therefore not treated as one generic group. An investigating police officer may request FIR details, case diary material, or evidence references for an assigned case. A ranked or supervisory police officer may approve, reject, or escalate requests that cross station, district, or sensitivity boundaries. A forensic expert or laboratory specialist may need controlled access to sample references, laboratory reports, or evidence-chain records. A prosecutor, court-linked authority, or auditor may need later access to verify whether a decision followed the correct policy and approval path. These roles motivate an access model that checks not only who the requester is, but also what record is requested, why it is requested, under which authority, and what evidence should remain for later review.

The problem addressed in this paper is therefore the following. Given a sensitive inter-agency request for a police or criminal-justice record, the system should decide whether the request must be allowed, denied, or escalated. At the same time, it should preserve evidence about the decision: the request context, the policy version, the attributes that mattered, the explanation given, and the

approval or review event connected to it. The design must avoid placing raw sensitive records on a shared ledger. It must also avoid assuming that one mechanism, whether blockchain, access control, or AI, is enough by itself.

### A. Practical Challenges

The paper is written around four practical challenges that arise when sensitive police or criminal-justice records are requested across agencies. Each challenge is paired with one contribution in the same order.

- Contextual access governance. Inter-agency police access cannot be handled only with static roles. Access depends on the requester, record, action, purpose, approval state, and operating context.
- Tamper-evident audit across agencies. A shared audit trail is useful only if later reviewers can trust that decision evidence has not been silently changed. At the same time, raw sensitive records should not be copied to a common blockchain.
- Explainable access decisions. When a police officer, investigating agency, court-related authority, or auditor requests access to a sensitive record, the system should explain why the request was allowed, denied, or escalated. This is important because such decisions can affect investigation timelines, privacy protection, inter-agency accountability, and later legal or administrative review.
- Valid-looking corrupted logs. A stronger attacker may change policy output and then re-sign the log. In that case, the log can look internally valid even though the decision has been corrupted.

### B. Research Contributions

To address these challenges, this paper makes the following four contributions. The ordering is intentional: Contribution 1 addresses contextual access governance, Contribution 2 addresses tamper-evident audit, Contribution 3 addresses explainable access decisions, and Contribution 4 addresses valid-looking corrupted logs.

- Contribution 1: The paper designs a layered access-control service for CCTNS/ICJS-compatible access requests. The service combines RBAC, ABAC, and PBAC-style rules and returns one of three decisions: allow, deny, or escalate.
- Contribution 2: The paper implements a permissioned blockchain-style audit layer that records request identifiers, policy versions, decision hashes, explanation hashes, approval references, record commitments, and block commitments while keeping raw records off-chain.
- Contribution 3: The paper builds an explainable decision service that records the decision label, reason code, decisive attributes, policy/rule version, counterfactual information where applicable, and an explanation hash linked to the audit event. The explanation is treated as an audit artifact, not only as a message shown on a screen.

- Contribution 4: The paper defines and runs a reproducible adversarial benchmark for ordinary tamper attacks and a validly re-signed compromised-signer attack. It compares mutable logs, signed hash chains, blockchain-style audit, CT-style logs, ABAC/Fabric-style re-execution, a trusted raw-attribute policy oracle, and NS-PI log-only drift detection.

The first contribution is mainly about security and access control. The second is about blockchain as an audit mechanism. The third is about explainable AI. The fourth makes the evaluation sharper by testing a failure mode where a log can remain cryptographically valid but semantically wrong. Together, these contributions keep the work focused on access governance rather than on broad claims about police automation.

The access-control layer follows the ABAC idea that decisions can be based on subject, object, action, and environmental attributes [10]. This fits the police data-sharing setting better than role-only access because the same role may require different decisions under different purposes, jurisdictions, and sensitivity levels. The blockchain layer is used narrowly: it gives tamper-evident audit commitments and provenance. It is not used as a privacy guarantee, and it does not store raw police records. This is consistent with the cautious treatment of blockchain access-control systems in NISTIR 8403 [11] and with the permissioned blockchain model represented by Hyperledger Fabric [12].

The explainable AI layer is also used in a narrow way. It does not explain crime prediction or suspect prediction. It explains access decisions. For each request, the system records why the request was allowed, denied, or escalated, which attributes were decisive, and what policy version was applied. This follows the broader argument that high-stakes systems should prefer interpretable and reviewable decisions when possible [13]. It also follows recent law-enforcement XAI work, which warns that explanations must support different stakeholders and should not create blind trust in automated systems [7].

The paper deliberately avoids the predictive-policing direction. Public National Crime Records Bureau data is useful for background context, but it is aggregate reported/registered crime data and is not a public individual-level CCTNS/FIR access-control dataset [14]. Prior work has also shown that predictive policing can create feedback loops when observed crime data is shaped by earlier policing decisions [15], and that complex criminal-justice prediction systems do not automatically outperform simple baselines [16]. For these reasons, this paper studies access governance, auditability, and explanation, not crime prediction.

The proposed system is named SEBA-XAI: a Secure, Explainable, Blockchain-Audited overlay for sensitive inter-agency access governance. The evaluation uses a synthetic workload because real police access logs and sensitive records are not publicly available. The claim is therefore controlled and limited. The paper does not claim live deployment inside CCTNS or ICJS, legal compliance, real police-record validation, production security, formal

privacy, or state-of-the-art performance. Its contribution is an implemented benchmark and prototype that studies how access control, blockchain-style audit, and XAI behave together under explicit trust and visibility assumptions, including baselines, ablations, and failure cases.

## II. Related Work

### A. Digital Policing and Inter-Agency Sharing

CCTNS and ICJS are the baseline systems for the Indian context. Official sources state that CCTNS is in use across all 17,798 police stations and that ICJS links the police/CCTNS pillar with courts, prisons, forensics, and prosecution [1], [2]. The public NCRB Crime in India data is useful for understanding registered crime at an aggregate level, but it is not an individual-level access-log or FIR dataset for this paper’s problem [14]. India-focused AI/crime studies use such public data for spatial or statistical analysis, but they do not model classified inter-agency access governance [17]–[19].

Recent legal research also warns that AI use in criminal justice raises data protection questions in India, especially because sensitive criminal-justice data can involve accused persons, suspects, victims, witnesses, and the wider public [6]. This supports the paper’s scope: the work does not replace CCTNS or ICJS, and it does not treat police records as ordinary open data. It asks what kind of secure overlay can sit above a CCTNS/ICJS-compatible environment when sensitive records must be requested, explained, and reviewed across agencies.

### B. Blockchain Audit and Evidence Management

Blockchain has already been studied for provenance, access control, and digital evidence management. NIS-TIR 8403 explains why blockchain can support tamper-resistant access-control records, while also noting practical concerns such as governance, scalability, and system design choices [11]. Hyperledger Fabric is a useful reference for a permissioned setting where the participants are known organizations rather than anonymous public miners [12]. Certificate Transparency also provides a non-blockchain reference point for append-only Merkle-tree audit logs and consistency proofs [20]. This is important because the paper does not assume that a blockchain is automatically better than every signed log.

Digital evidence systems and lawful-evidence chains show how blockchains can support chain-of-custody, provenance, off-chain evidence storage, and forensic integrity [21]–[23]. Other work studies blockchain-based auditable access control, privacy-preserving access control, forensic log access, and digital evidence protection [3], [5], [24]–[26]. Blockchain has also been used in smart public-safety systems where distributed participants need traceability and auditability [27].

These works are relevant, but they do not fully solve the problem studied in this paper. Evidence provenance is not the same as deciding whether a sensitive inter-agency

access request should be allowed, denied, or escalated. A ledger can show that a record of a decision was not changed, but it cannot by itself prove that the decision was correct, privacy-preserving, or properly explained.

### C. Access Control for Sensitive Records

RBAC is a mature baseline for organizing permissions around roles [28]. However, role-only access is too coarse for the setting studied here. ABAC allows authorization to depend on subject, object, action, and environmental attributes [10]. XACML is also important as a policy-language reference because it formalizes policy decisions, obligations, advice, and policy evaluation structures [29]. For police data sharing, access may depend on jurisdiction, purpose, record sensitivity, approval status, credential validity, case assignment, and emergency context.

Fabric-based ABAC work shows that blockchain infrastructure and access control can be combined with off-chain encrypted storage patterns [30]. Privacy-preserving ABAC work shows that even attribute disclosure during authorization can create privacy concerns, motivating selective-disclosure, homomorphic-encryption, and zero-knowledge approaches [4]. The present paper does not implement a full cryptographic privacy-preserving ABAC protocol. Instead, it treats metadata exposure as an evaluation issue and keeps raw sensitive records off-chain.

However, generic ABAC or blockchain-ABAC work usually does not include a police-specific allow/deny/escalate workflow. It also does not usually treat the explanation itself as an artifact that must be logged and checked later. This is the gap used in this paper: access control is necessary, but it must be connected to audit evidence and explanation evidence.

### D. Explainable AI and High-Stakes Review

The XAI part of this paper is not added for presentation. It is part of the audit model. Rudin argues that high-stakes decisions should prefer interpretable models where this is feasible [13]. Recent work on law-enforcement XAI also highlights that explanations must support different stakeholders and must not encourage blind trust in automated systems [7]. Work on human-centered explanation shows that explanations are social and contrastive, not just feature lists [31]. Counterfactual explanations are also relevant because an access requester may need to know what missing condition would have changed a denial or escalation decision [32].

Standard XAI methods such as LIME and SHAP explain model predictions through local surrogate explanations and feature attributions [33], [34]. This paper does not depend on a black-box predictor for the main access decision; it uses a deterministic policy oracle so that labels are transparent in the first benchmark. Still, the XAI literature is useful for defining explanation artifacts, explanation coverage, counterfactual validity, and evaluation discipline [35].

There is also work that directly connects blockchain and XAI. Nassar et al. propose blockchain support for trustworthy and explainable AI systems [36]; Malhotra et al. study blockchain audit trailing of XAI decisions using IPFS and Ethereum [37]; and legal/XAI studies discuss blockchain-based audit of legal or justice-related AI decisions [38], [39]. These works support the idea of logging explanation evidence, but they do not provide the CCTNS/ICJS-compatible police access-governance benchmark studied here.

#### E. Why This Is Not Predictive Policing

This paper avoids individual crime prediction and suspect prediction. Prior work has shown that predictive-policing systems can create feedback loops when future data is influenced by earlier policing activity [15]. Algorithmic policing work also emphasizes assumptions, evaluation, and accountability problems in prediction-led policing [40]. Machine-learning crime prediction has a large literature, but the datasets, labels, and evaluation conditions vary widely [41]. Criminal-justice prediction work also shows that complex models are not automatically better than simpler baselines [16]. Fairness literature shows that some fairness criteria cannot all be satisfied simultaneously under common conditions [42], [43].

These findings support a cautious framing. The paper uses AI to explain access decisions and detect policy drift in logs, not to predict criminals or crime locations.

#### F. Research Gap

The existing literature gives useful building blocks: blockchain audit, permissioned ledgers, ABAC, off-chain storage, and XAI for high-stakes settings. The missing piece is their joint evaluation in an inter-agency access governance workflow for CCTNS/ICJS-compatible police record requests. In particular, the literature does not sufficiently study what happens when a log is validly re-signed after policy output is corrupted. In that case, ordinary integrity checks may pass even though the decision is wrong. This paper studies that gap with a synthetic benchmark and compares defenses under clear visibility assumptions.

### III. Proposed Methodology

#### A. System View

The proposed system is a research prototype and benchmark. It is not presented as a production deployment. The design has four main parts: an access request gateway, a layered access-control service, an explainable decision service, and a permissioned blockchain-style audit layer. Raw records remain in agency-controlled off-chain storage. The audit layer stores only identifiers, commitments, hashes, policy versions, approval references, explanation hashes, and block metadata.

#### B. Access Request Flow

The workflow begins when a user requests a sensitive record. The user may be a police officer, an investigating agency user, a forensic user, a prosecutor or court-related authority, or an auditor. The gateway converts the request into a structured form. Subject attributes include role, rank, station, jurisdiction, credential status, and case assignment. Object attributes include record type, sensitivity level, juvenile or witness flags, sealed status, and owning agency. Action attributes describe what the requester wants to do. Environment attributes include purpose, time window, emergency flag, court/prosecutor link, approval token, and policy version.

The access-control service evaluates these attributes and returns allow, deny, or escalate. A request can be allowed when the requester has the required role, assignment, purpose, and approval context. It can be denied when the credential is revoked, the approval is invalid, or the request clearly violates policy. It can be escalated when a superior or review authority must check the request, such as in cross-jurisdiction or sensitive-record cases.

#### C. Policy Layer

The policy layer combines RBAC, ABAC, and PBAC-style rules. RBAC gives the basic user categories. ABAC adds contextual attributes such as jurisdiction, case assignment, purpose, sensitivity, and credential state. PBAC adds policy versioning, approval requirements, revocation handling, and exceptional cases. The current prototype uses a deterministic policy oracle because the first paper needs transparent benchmark labels. This is more suitable than a black box model for the initial access decision.

#### D. Explainable Decision Layer

After a decision is made, the XAI layer creates an explanation artifact. The artifact contains the decision label, reason code, decisive attributes, policy or rule version, counterfactual information where applicable, and an explanation hash. A typical explanation answers a direct question: why was this request allowed, denied, or escalated?

This layer is designed for review, not only for display. A supervisor can use it to inspect an escalation. An auditor can use it to check whether the policy and explanation match. A court-related or administrative reviewer can use it to understand what information was considered at the time of access. The explanation hash is linked to the audit event so that a later change in the explanation can be detected.

#### E. Blockchain-Style Audit Layer

The audit layer is modeled as a permissioned blockchain-style ledger among known agencies. Each audit event stores minimized metadata and commitments rather than raw sensitive records. A block contains a batch of events, a previous-block hash, an event commitment,

validator identity, and signature. The design is compared with simpler baselines such as a mutable log and a signed append-only hash chain. This comparison is important because blockchain should not be assumed to help unless it is tested against simpler audit mechanisms.

#### F. Synthetic Benchmark

Real police access logs are not publicly available, and they should not be assumed in a first-stage academic prototype. The benchmark therefore generates synthetic stations, officers, cases, records, and access requests. The declared policy oracle labels each request as allow, deny, or escalate. The synthetic setting gives control over request context, policy stress cases, and attack conditions. It does not claim to represent real CCTNS or ICJS traffic.

#### G. Threat Model and Evaluation

The attack catalog includes ordinary tampering, approval-token replay, request backdating, explanation-hash substitution, revocation-race behavior, metadata-inference checks, block-signature collusion, and the validly re-signed compromised-signer attack. The compromised-signer case is important because an attacker may corrupt the policy output and then produce a log that still looks properly signed. This separates integrity of the log from correctness of the decision.

The evaluation compares mutable logs, signed hash chains, blockchain-style audit, CT-style logs, ABAC/Fabric-style re-execution, a trusted raw-attribute policy oracle, and NS-PI log-only drift detection. The research questions are:

- RQ1: Which audit designs detect ordinary tamper attacks?
- RQ2: Which defenses detect validly re-signed policy corruption?
- RQ3: What are the sensitivity limits of NS-PI drift detection?
- RQ4: How complete and stable are the XAI artifacts and audit reconstruction traces?
- RQ5: What metadata exposure and local latency/storage overhead are introduced by the audit design?

The metrics include adversarial audit score, per-attack detection rate, trace-completeness rate, decisive-attribute coverage, counterfactual coverage, counterfactual validity, explanation stability, audit reconstruction rate, metadata exposure score, decision/build latency, verification latency, and storage overhead. The experiments are run with deterministic seeds {7, 21, 42, 99, 123}. The paper reports results only from generated artifacts in the repository and does not infer real police-system performance from the synthetic workload.

### IV. Results and Discussion

#### A. Experimental Scope

The results in this section come from the synthetic SEBA-XAI benchmark and should be read only within

TABLE I  
Selected method comparison on the 1,000-request evidence workload

| Method                       | Acc.   | Tamper | XAI hash | p50 ms |
|------------------------------|--------|--------|----------|--------|
| RBAC + mutable log           | 0.2900 | 0.0000 | No       | 7.54   |
| ABAC/PBAC + mutable log      | 1.0000 | 0.0000 | No       | 20.99  |
| ABAC/PBAC + signed chain     | 1.0000 | 1.0000 | No       | 30.99  |
| ABAC/PBAC + blockchain audit | 1.0000 | 1.0000 | No       | 42.09  |
| SEBA-XAI full                | 1.0000 | 1.0000 | Yes      | 42.09  |

that scope. The workload contains generated stations, officers, records, policy contexts, and access requests. The benchmark is useful for testing whether the architecture is measurable and whether different audit designs fail in different ways. It is not evidence of performance on live CCTNS or ICJS data.

#### B. Baseline Access-Control and Audit Comparison

Table I shows the main prototype comparison on the 1,000-request paper evidence workload. A role/action-only RBAC baseline reaches 0.2900 accuracy against the declared policy oracle and creates 656 false allows. This supports the design choice that role-only access is too weak for the synthetic sensitive-record workload. The configured ABAC/PBAC policy matches the declared oracle in this benchmark. However, the mutable-log version does not provide internal tamper evidence. Adding a signed hash chain or blockchain-style audit gives tamper evidence for controlled ordinary edits, while the full SEBA-XAI mode additionally logs the explanation hash.

The interpretation is not that blockchain is always better than a signed hash chain. The local result only shows that both integrity-preserving append-only designs detect the controlled ordinary tamper cases. The reason to keep the blockchain-style layer in the architecture is the inter-agency audit setting: known agencies can maintain a shared audit commitment without placing raw records on-chain.

#### C. Compromised-Signer Result

The strongest technical result is the compromised-signer asymmetry. In the five-seed adversarial grid, mutable logs, signed chains, blockchain-style audit, CT-style logs, ABAC re-execution, and Fabric-style ABAC all record a 0.0000 detection rate for the validly re-signed compromised-signer attack. NS-PI drift and the trusted raw-attribute policy oracle record a 1.0000 detection rate for that attack. This result does not mean NS-PI is the best overall defense. Its mean adversarial audit score is 0.2500, while the signed chain, blockchain-style audit, CT-style log, ABAC re-execution, and Fabric-style ABAC each score 0.7917 across the broader attack set. The trusted policy oracle scores 1.0000 because it has the strongest visibility assumption: access to an uncompromised raw request view.

This is the main reason the paper treats the system as layered. Integrity mechanisms answer whether the recorded log was changed. Policy re-evaluation answers

TABLE II  
Five-seed audit results for selected defenses

| Defense                        | AAS mean | Compromised signer |
|--------------------------------|----------|--------------------|
| Mutable log                    | 0.5000   | 0.0000             |
| Signed hash chain              | 0.7917   | 0.0000             |
| Blockchain-style audit         | 0.7917   | 0.0000             |
| CT-style log                   | 0.7917   | 0.0000             |
| ABAC/Fabric-style re-execution | 0.7917   | 0.0000             |
| NS-PI drift                    | 0.2500   | 1.0000             |
| Trusted policy oracle          | 1.0000   | 1.0000             |

whether the recorded decision agrees with policy under the available attributes. NS-PI answers a narrower log-only question: whether the signed decision trace has shifted in a suspicious way. These are different questions, so the defenses should not be treated as interchangeable.

#### D. NS-PI Sensitivity and Workload Stress

The NS-PI results are useful but limited. In global compromised-signer sensitivity tests, NS-PI misses 2% and 5% corruption but detects the attack from 10% corruption onward in the current workload. For targeted attacks, global drift can miss localized corruption because the total changed fraction of the whole workload is small. Grouped station/district drift is more useful: station-level grouped NS-PI reaches full detection from 50% of targeted eligible station rows, and district-level grouped NS-PI reaches full detection from 25% of targeted eligible district rows. Very small 10% targeted corruption is still missed.

Workload stress tests over 500, 1,000, 2,500, and 5,000 generated requests show the same basic compromised-signer pattern at 25% corruption: signed chains remain at 0.0000 detection, while NS-PI global detection and the trusted oracle remain at 1.0000. At 10% corruption, the signal is weaker for smaller workloads; for example, the 500-request baseline arm has 0.2000 NS-PI global detection across five seeds, while the 1,000-request and larger baseline arms reach 1.0000. This is a useful boundary condition for the paper: NS-PI needs enough traffic and enough corruption signal to be reliable.

#### E. XAI and Audit Reconstruction

The XAI component is strongest as a structured trace. Across five seeds, trace completeness is 1.0000, counterfactual coverage is 1.0000, counterfactual validity is 0.9964, stable decision/reason rate is 1.0000, and audit reconstruction rate is 1.0000. These results support the claim that the prototype can produce reviewable decision artifacts and link them back to audit events.

The natural-language explanation is weaker than the structured trace. The mean decisive-attribute text coverage is 0.9310, but the full text coverage rate is 0.7810. Therefore the paper should not claim that the explanation text is complete. A fair claim is that the structured XAI artifact is complete in this benchmark, while the text renderer still needs improvement before stronger human-facing explanation claims are made.

#### F. Metadata, Latency, and Storage

The metadata experiment compares a full metadata ledger with a minimized commitment ledger. The full metadata ledger has a schema-level exposure score of 1.0000 because fields such as decision, purpose, requester station, and target station remain visible. The minimized commitment ledger has a score of 0.0000 in the same proxy because sensitive context is represented by hashes or commitments. This is not a formal privacy proof; it only shows that schema minimization reduces direct metadata exposure in the prototype.

The local latency/storage measurements are also modest-scope prototype measurements. Policy-oracle and XAI generation takes 14.46 ms p50 for 1,000 requests in the benchmark artifact. Mutable-log build time is 6.84 ms p50, signed-chain build and verification are 16.84 ms and 9.16 ms p50, and permissioned blockchain-style build and verification are 11.10 ms and 2.50 ms p50 in the local simulation. Storage per event is 459.82 bytes for the mutable log, 756.86 bytes for the signed hash chain, and 353.50 bytes for the blockchain-style audit artifact used in this measurement. These numbers should not be interpreted as production throughput, but they show that the prototype records latency and storage overhead in a reproducible way.

#### V. Limitations

The current paper should be read as a synthetic benchmark and prototype study. It does not use real FIR records, CCTNS logs, ICJS logs, police access logs, or sensitive operational data. The policy oracle is a declared benchmark policy, not an official police policy. Therefore, the results show whether the proposed architecture is technically measurable, not whether it is ready for direct deployment.

The blockchain layer is also limited. It is a local permissioned audit simulation, not a live Hyperledger Fabric deployment. The metadata-exposure result is a schema-level proxy, not a formal privacy proof. The audit layer can support tamper evidence and reconstruction, but it does not prove that a policy decision was correct.

NS-PI has a narrow role. It is not the best overall tamper detector and it does not replace cryptographic audit, ABAC/PBAC enforcement, or trusted raw-attribute policy checking. It is useful as a complementary log-only drift signal for validly re-signed compromised-signer behavior. It still misses low-rate and small targeted attacks in the current benchmark, especially 2%–5% global corruption and 10% targeted station/district corruption.

The XAI layer is strong as a structured trace, but it is not perfect as natural language. The benchmark records decision labels, reason codes, decisive attributes, policy versions, counterfactual information, and explanation hashes. However, the rendered explanation text does not always mention every decisive attribute. This should be improved before the paper makes stronger claims about explanation quality.

Any real-world pilot would require official policy validation, legal and privacy review, secure integration design, institutional approval, and testing with appropriate governance. This paper does not claim those outcomes.

## VI. Conclusion

This paper presents a secure, explainable, blockchain-audited overlay for inter-agency police record access governance. The main idea is simple: sensitive record sharing needs contextual access control, tamper-evident audit, and explanations that can be reviewed later. None of these parts is sufficient alone.

The proposed work contributes a layered access-control service for allow/deny/escalate decisions, a permissioned blockchain-style audit layer for commitments and decision evidence, and an explainable decision service that records why each access request was allowed, denied, or escalated. The benchmark also studies a validly re-signed compromised-signer attack, where a log can appear valid even after policy output is corrupted.

The work remains intentionally bounded. It is a synthetic benchmark and research prototype, not an operational police deployment. The results support a layered interpretation: blockchain-style audit improves tamper evidence for recorded events, contextual policy checking improves decision accountability, and explainable drift/audit artifacts help review failure modes that are not visible to a single mechanism alone. Future work should validate policies with domain experts, improve the natural-language explanation renderer, strengthen privacy analysis, and test the design under officially approved operational constraints.

## References

- [1] Press Information Bureau, Government of India, "CCTNS operational police stations," Mar. 2026, accessed: May 30, 2026. [Online]. Available: <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2238241>
- [2] Ministry of Home Affairs, Government of India, "ICJS/NCRB administration," 2026, accessed: May 30, 2026. [Online]. Available: <https://www.mha.gov.in/en/commoncontent/icjsncrb-administration>
- [3] Q. Hu, C. Huang, G. Zhang, L. Cai, and T. Jiang, "Towards accountable and privacy-preserving blockchain-based access control for data sharing," *Journal of Information Security and Applications*, vol. 85, p. 103866, 2024.
- [4] M. Kerl, U. Bodin, and O. Schelén, "Privacy-preserving attribute-based access control using homomorphic encryption," *Cybersecurity*, vol. 8, no. 5, 2025.
- [5] J. Banerjee, S. Dhoundiyal, M. Ghosh, R. Raj, and A. Wary, "dashdikpala: A Blockchain-Enabled Lattice Based Access Control Scheme for Digital Evidence Data Protection," *Peer-to-Peer Networking and Applications*, vol. 19, no. 46, 2026.
- [6] G. Anbarasi and D. Sankar, "Data Protection Challenges in AI-Driven Criminal Justice in the EU and India," *International Journal of Computational Intelligence Systems*, vol. 19, no. 20, 2026.
- [7] M. Zocholl, D. Stampouli, M. Wittfoth, and G. Mounier, "Fundamental considerations for the use of explainable AI in law enforcement," *Frontiers in Political Science*, vol. 7, p. 1605619, 2025.
- [8] The Guardian, "Northern Ireland police officers' details exposed in 'monumental' breach," Aug. 2023, accessed: June 25, 2026. [Online]. Available: <https://www.theguardian.com/uk-news/2023/aug/08/major-data-breach-involving-northern-ireland-police-officers-and-staff>
- [9] Information Commissioner's Office, "What price privacy? Poor PSNI procedures culminate in £750k fine," Oct. 2024, accessed: June 25, 2026. [Online]. Available: <https://ico.org.uk/about-the-ico/media-centre/news-and-blogs/2024/10/what-price-privacy-poor-psni-procedures-culminate-in-750k-fine/>
- [10] National Institute of Standards and Technology, "Guide to Attribute Based Access Control (ABAC) Definition and Considerations," National Institute of Standards and Technology, Tech. Rep. SP 800-162, 2014. [Online]. Available: <https://csrc.nist.gov/pubs/sp/800/162/upd2/final>
- [11] —, "Blockchain for Access Control Systems," National Institute of Standards and Technology, Tech. Rep. NISTIR 8403, 2022.
- [12] E. Androulaki, A. Barger, V. Bortnikov, C. Cachin, K. Christidis, A. D. Caro, D. Enyeart, C. Ferris, G. Laventman, Y. Manevich, S. Muralidharan, C. Murthy, B. Nguyen, M. Sethi, G. Singh, K. Smith, A. Sorniotti, C. Stathakopoulou, M. Vukolić, S. W. Cocco, and J. Yellick, "Hyperledger Fabric: A Distributed Operating System for Permissioned Blockchains," in *Proc. Thirteenth EuroSys Conference*, 2018, pp. 30:1–30:15.
- [13] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, 2019.
- [14] National Crime Records Bureau / data.gov.in, "Crime in India 2023," 2023, accessed: May 30, 2026. [Online]. Available: <https://www.data.gov.in/catalog/crime-india-2023>
- [15] D. Ensign, S. A. Friedler, R. Neville, C. Scheidegger, and S. Venkatasubramanian, "Runaway Feedback Loops in Predictive Policing," in *Proc. FAT*, 2018. [Online]. Available: <https://proceedings.mlr.press/v81/ensign18a.html>
- [16] J. Dressel and H. Farid, "The accuracy, fairness, and limits of predicting recidivism," *Science Advances*, 2018.
- [17] R. Sharma and U. Gupta, "Analysis of criminal spatial events in India using exploratory data analysis and regression," *Computers and Electrical Engineering*, vol. 109, p. 108761, 2023.
- [18] A. S. Naick, R. C. Poonia, and A. Sharma, "Forecasting Murder Motives in India Using Statistical Analysis and Explainable Artificial Intelligence," in *Proc. IEEE Uttar Pradesh Section International Conference on Electrical, Electronics and Computer Engineering (UPCON)*, 2024, pp. 1–6.
- [19] M. A. Reza, A. Bisaria, S. Advaita, A. Ponnekanti, and A. Arya, "CriX: Intersection of Crime, Demographics and Explainable AI," in *Proc. 17th International Conference on Agents and Artificial Intelligence (ICAART)*, vol. 2, 2025, pp. 714–725.
- [20] B. Laurie, A. Langley, and E. Kasper, "Certificate Transparency," RFC 6962, Jun. 2013. [Online]. Available: <https://www.rfc-editor.org/rfc/rfc6962.html>
- [21] D. Kim, S.-Y. Ihm, and Y. Son, "Two-Level Blockchain System for Digital Crime Evidence Management," *Sensors*, vol. 21, no. 9, p. 3051, 2021.
- [22] M. Li, C. Lal, M. Conti, and D. Hu, "LEChain: A blockchain-based lawful evidence management scheme for digital forensics," *Future Generation Computer Systems*, vol. 115, pp. 406–420, 2021.
- [23] J. Jeong, D. Kim, B. Lee, and Y. Son, "Design and Implementation of a Digital Evidence Management Model Based on Hyperledger Fabric," *Journal of Information Processing Systems*, vol. 16, no. 4, pp. 760–773, 2020.
- [24] A. Akhtar, B. Shafiq, J. Vaidya, A. Afzal, S. Shamail, and O. Rana, "Blockchain Based Auditable Access Control for Distributed Business Processes," in *Proc. IEEE 40th International Conference on Distributed Computing Systems (ICDCS)*, 2020, pp. 12–22.
- [25] S. Rouhani and R. Deters, "Blockchain based access control systems: State of the art and challenges," 2019. [Online]. Available: <https://arxiv.org/abs/1908.08503>
- [26] M. E. Islam, M. R. Islam, M. Chetty, S. Lim, and M. Chadhar, "User authentication and access control to blockchain-based forensic log data," *EURASIP Journal on Information Security*, vol. 2023, no. 7, 2023.
- [27] R. Xu, S. Y. Nikouei, D. Nagothu, A. Fitwi, and Y. Chen, "BlendSPS: A BLockchain-ENabled Decentralized Smart Public Safety System," *Smart Cities*, vol. 3, no. 3, pp. 928–951, 2020.
- [28] R. S. Sandhu, E. J. Coyne, H. L. Feinstein, and C. E. Youman, "Role-Based Access Control Models," *IEEE Computer*, vol. 29, no. 2, pp. 38–47, 1996.

- [29] OASIS, “eXtensible Access Control Markup Language (XACML) Version 3.0,” Jan. 2013, oASIS Standard, 22 January 2013. [Online]. Available: <https://docs.oasis-open.org/xacml/3.0/xacml-3.0-core-spec-os-en.html>
- [30] X. Zhao, S. Wang, Y. Zhang, and Y. Wang, “Attribute-based access control scheme for data sharing on Hyperledger Fabric,” *Journal of Information Security and Applications*, vol. 67, p. 103182, 2022.
- [31] T. Miller, “Explanation in artificial intelligence: Insights from the social sciences,” *Artificial Intelligence*, vol. 267, pp. 1–38, 2019.
- [32] S. Wachter, B. Mittelstadt, and C. Russell, “Counterfactual Explanations Without Opening the Black Box: Automated Decisions and the GDPR,” *Harvard Journal of Law & Technology*, vol. 31, no. 2, pp. 841–887, 2018. [Online]. Available: <https://jolt.law.harvard.edu/assets/articlePDFs/v31/Counterfactual-Explanations-without-Opening-the-Black-Box-Sandra-Wachter-et-al.pdf>
- [33] M. T. Ribeiro, S. Singh, and C. Guestrin, ““Why Should I Trust You?”: Explaining the Predictions of Any Classifier,” in *Proc. 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 1135–1144.
- [34] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: [https://papers.nips.cc/paper\\_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html](https://papers.nips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html)
- [35] F. Doshi-Velez and B. Kim, “Towards A Rigorous Science of Interpretable Machine Learning,” 2017. [Online]. Available: <https://arxiv.org/abs/1702.08608>
- [36] M. Nassar, K. Salah, M. H. U. Rehman, and D. Svetinovic, “Blockchain for explainable and trustworthy artificial intelligence,” *WIREs Data Mining and Knowledge Discovery*, vol. 10, no. 1, p. e1340, 2020.
- [37] D. Malhotra, S. Srivastava, P. Saini, and A. K. Singh, “Blockchain based audit trailing of XAI decisions: Storing on IPFS and Ethereum Blockchain,” in *Proc. International Conference on Communication Systems & NETWORKS (COMSNETS)*, 2021, pp. 1–5.
- [38] S. Sachan and X. Liu, “Blockchain-based auditing of legal decisions supported by explainable AI and generative AI tools,” *Engineering Applications of Artificial Intelligence*, vol. 129, p. 107666, 2024.
- [39] K. Demertzis, K. Rantos, L. Magafas, C. Skianis, and L. Iliadis, “A Secure and Privacy-Preserving Blockchain-Based XAI-Justice System,” *Information*, vol. 14, no. 9, p. 477, 2023.
- [40] L. B. Moses and J. Chan, “Algorithmic prediction in policing: assumptions, evaluation, and accountability,” *Policing and Society*, vol. 28, no. 7, pp. 806–822, 2018.
- [41] K. Jenga, C. Catal, and G. Kar, “Machine learning in crime prediction,” *Journal of Ambient Intelligence and Humanized Computing*, vol. 14, no. 3, pp. 2887–2913, 2023.
- [42] A. Chouldechova, “Fair prediction with disparate impact: A study of bias in recidivism prediction instruments,” *Big Data*, vol. 5, no. 2, pp. 153–163, 2017.
- [43] J. Kleinberg, S. Mullainathan, and M. Raghavan, “Inherent Trade-Offs in the Fair Determination of Risk Scores,” 2016. [Online]. Available: <https://arxiv.org/abs/1609.05807>